Flight Performance & Analytics

Delay Causes & Pattern Analysis

Comprehensive Flight Delay & Regional Performance Report

An in-depth evaluation of departure and arrival delay metrics, carrier impact analysis, and key operational insights across major airports.



SkyWest Airlines: Exploring Flight Operations and On-Time Performance in 2025

An interactive data visualization study of flight volume, delay patterns, airport performance, and competitor comparison.



2025
Study Period

12 MONTHS
Data Coverage

OO
SkyWest Carrier Code

RESEARCH FOCUS

01 Flight Operations

Flight volume, monthly activity, and airport distribution.

02 On-Time Performance

Departure and arrival delays, delay patterns, and delay causes.

03 Competitor Comparison

SkyWest performance compared with selected regional carriers.

PRESENTATION ROADMAP

01 Data Preparation › 02 SkyWest Overview › 03 Performance Analysis › 04 Competitor Comparison

DATA COLLECTION

SOURCE	**Bureau of Transportation Statistics (BTS)** U.S. Airline On-Time Performance data
DATA COVERAGE	**January - December 2025** 12 monthly files SkyWest Airlines Inc. (OO)
DATASET SIZE	**839,821** SkyWest flight records
DATA CONTENT	Flight dates, origin and destination airports, flight times, delays, cancellations, diversions, flight distance, and delay causes.
PROJECT FOCUS	The data supports analysis of flight volume, airport activity, route performance, delays, delay causes, and competitor comparison.



Important Fields / Data Dictionary

Time Period

FIELD	DESCRIPTION
YEAR	Year
MONTH	Month
DAY_OF_WEEK	Day of Week
FL_DATE	Flight Date

Airline & Route

FIELD	DESCRIPTION
OP_UNIQUE_CARRIER	Unique airline carrier code
OP_CARRIER_FL_NUM	Flight number
ORIGIN	Origin airport
DEST	Destination airport
ORIGIN_CITY_NAME	Origin city
DEST_CITY_NAME	Destination city

Performance Metrics

FIELD	DESCRIPTION
DEP_DELAY	Departure delay (minutes)
ARR_DELAY	Arrival delay (minutes)
DEP_DEL15	Departure delay ≥ 15 min
ARR_DEL15	Arrival delay ≥ 15 min
DISTANCE	Flight distance (miles)
FLIGHTS	Flight volume (number of flights)

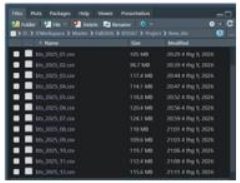
Operations & Delay Causes

FIELD	DESCRIPTION
CANCELLED	Cancelled flight indicator
DIVERTED	Diverted flight indicator
CARRIER_DELAY	Carrier delay (minutes)
WEATHER_DELAY	Weather delay (minutes)
NAS_DELAY	NAS delay (minutes)
SECURITY_DELAY	Security delay (minutes)
LATE_AIRCRAFT_DELAY	Late aircraft delay (minutes)

DATA PREPARATION SUMMARY

01 Source Files

12 monthly BTS CSV files
January - December 2025



02 Union

Read each monthly CSV
Combined using readcsv() + rbind()

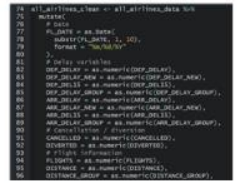
```
files <- c(
  "bts_2025_01.csv", "bts_2025_02.csv", "bts_2025_03.csv", "bts_2025_04.csv",
  "bts_2025_05.csv", "bts_2025_06.csv", "bts_2025_07.csv", "bts_2025_08.csv",
  "bts_2025_09.csv", "bts_2025_10.csv", "bts_2025_11.csv", "bts_2025_12.csv"
)
all_airlines_data <- data.frame()

for (f in files) {

  temp <- read.csv(
    paste0(
      "D:/ITWorkspace/Master/Fall2026/IDS567/Project/New_bts/",
      f
    )
  )
  all_airlines_data <- rbind(
    all_airlines_data,
    temp
  )
}
```

03 Data Cleaning

Standardize dates and numeric fields



04 Caculated Fields

Created 03 caculated fields
- ROUTE
- ON_TIME_DEP
- ON_TIME_ARR

```
  # Route
  ROUTE = paste(
    ORIGIN,
    DEST,
    sep = "-"
  ),
  # On-time indicators
  ON_TIME_DEP = ifelse(
    CANCELLED == 1,
    NA,
    ifelse(DEP_DEL15 == 0, 1, 0)
  ),
  ON_TIME_ARR = ifelse(
    CANCELLED == 1,
    NA,
    ifelse(ARR_DEL15 == 0, 1, 0)
  )
)
```

05 Data Quality Checks

Checked
- Missing Values
- Cancellation & Diversion status
- Delay Status
- Date coverage
- Duplicate rows

```
# Missing values
colSums(is.na(all_airlines_clean))
# Cancellation / diversion
table(all_airlines_clean$CANCELLED)
table(all_airlines_clean$DIVERTED)
# Delay status
table(all_airlines_clean$DEP_DEL15, useNA = "ifany")
table(all_airlines_clean$ARR_DEL15, useNA = "ifany")
# Date Coverage
table(factor(all_airlines_clean$MONTH, levels = 1:12))
# Duplicate
sum(duplicated(all_airlines_clean))
```

06 SkyWest Filter

Selected SkyWest Airlines Inc. (OO)

```
skywest_clean <- all_airlines_clean %>%
  filter(
    OP_UNIQUE_CARRIER == "OO"
  )
```

Final Dataset

839,821
SkyWest flight records

2025 Year	January - December Coverage
OO SkyWest Airlines Inc.	41 Total Fields

0
Duplicate Rows

```
Console  Terminal  Background Jobs
R 4.6.1 · D:/ITWorkspace/Master/Fall2026/IDS567/Project/
> skywest_clean <- read.csv(
+   "D:/ITWorkspace/Master/Fall2026/IDS567/Project/New_bts/only_skywest_2025_cleaned.csv"
+ )
> nrow(skywest_clean)
[1] 839821
> ncol(skywest_clean)
[1] 41
> table(skywest_clean$OP_UNIQUE_CARRIER)

    OO
839821
> table(factor(skywest_clean$MONTH,levels = 1:12))

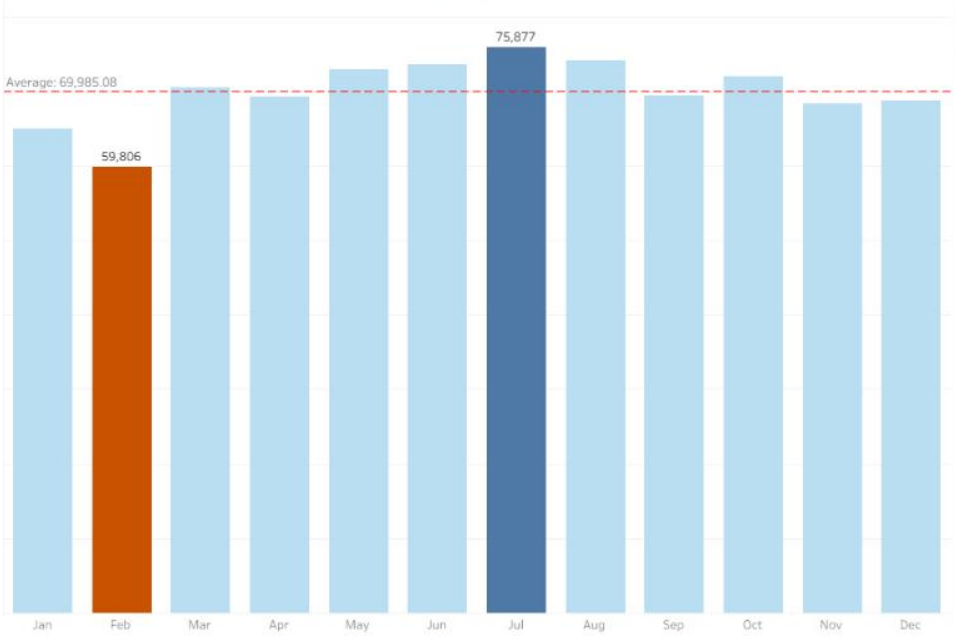
    1     2     3     4     5     6     7     8     9    10    11    12
65036 59806 70517 69319 72962 73660 75877 74150 69406 71927 68347 68814
> sum(skywest_clean$FLIGHTS)
[1] 839821
> sum(duplicated(skywest_clean))
[1] 0
> min(skywest_clean$FL_DATE, na.rm = TRUE)
[1] "2025-01-01"
> max(skywest_clean$FL_DATE, na.rm = TRUE)
[1] "2025-12-31"
```

SKYWEST AIRLINES - 2025 KPI OVERVIEW

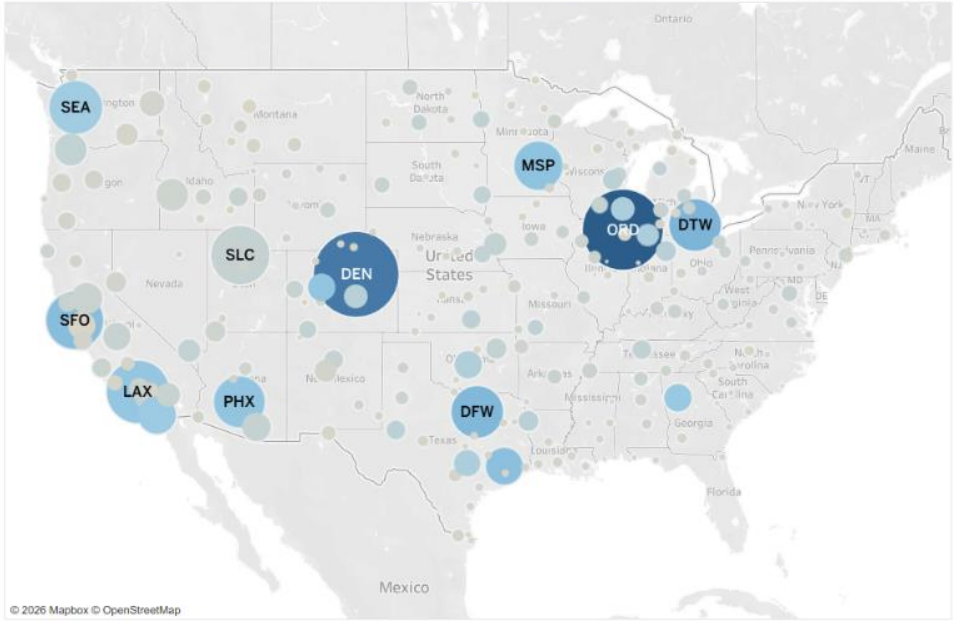
Flight volume and overall operational performance

839,821	12.67 min	9.15 min	80.41%	1.41%
Total Flights	Avg Departure Delay	Avg Arrival Delay	On-Time Performance	Cancellation Rate

Monthly Flight Volume



Origin Airport Activity

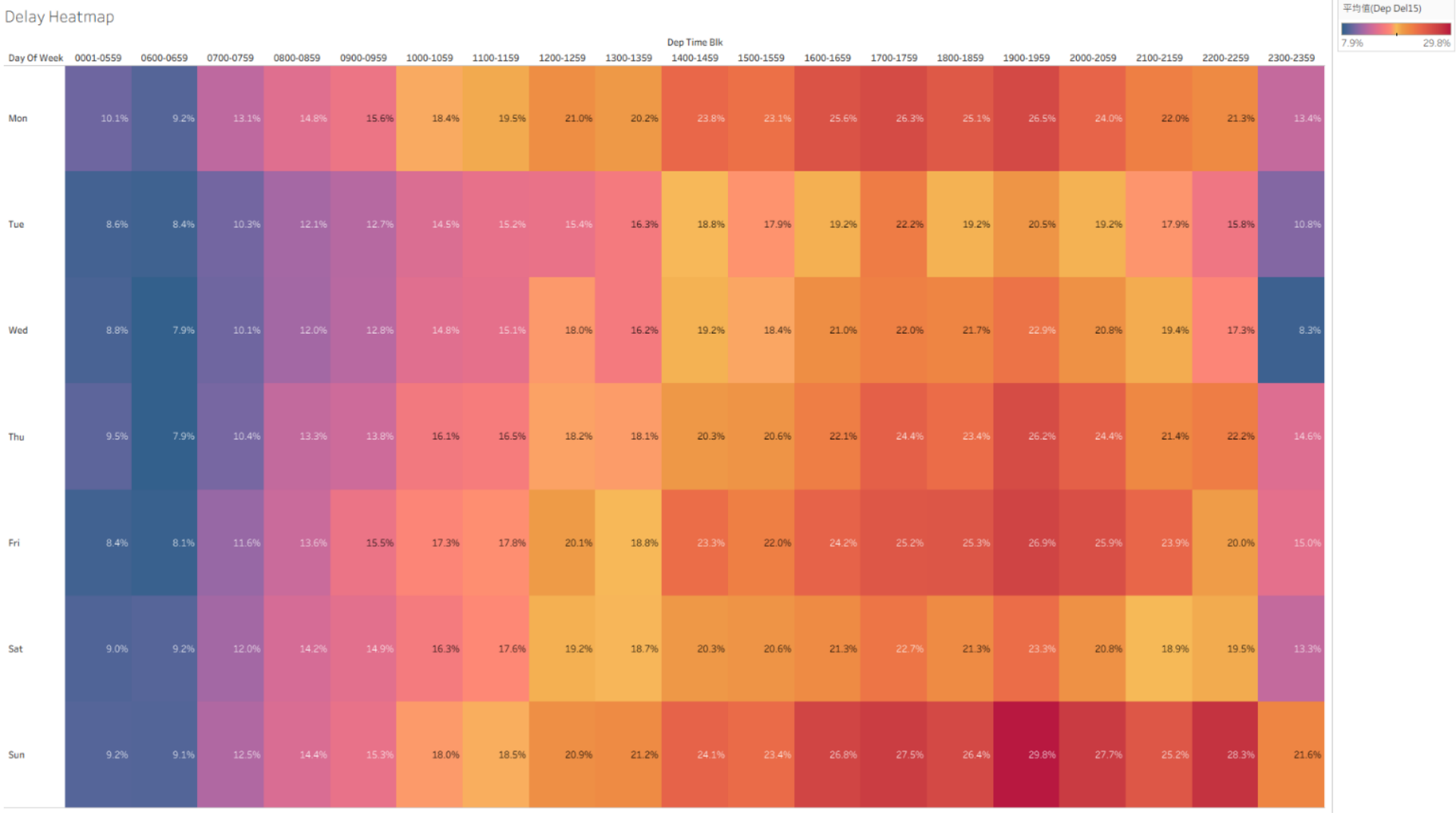


Monthly Departure & Arrival Delay Rate

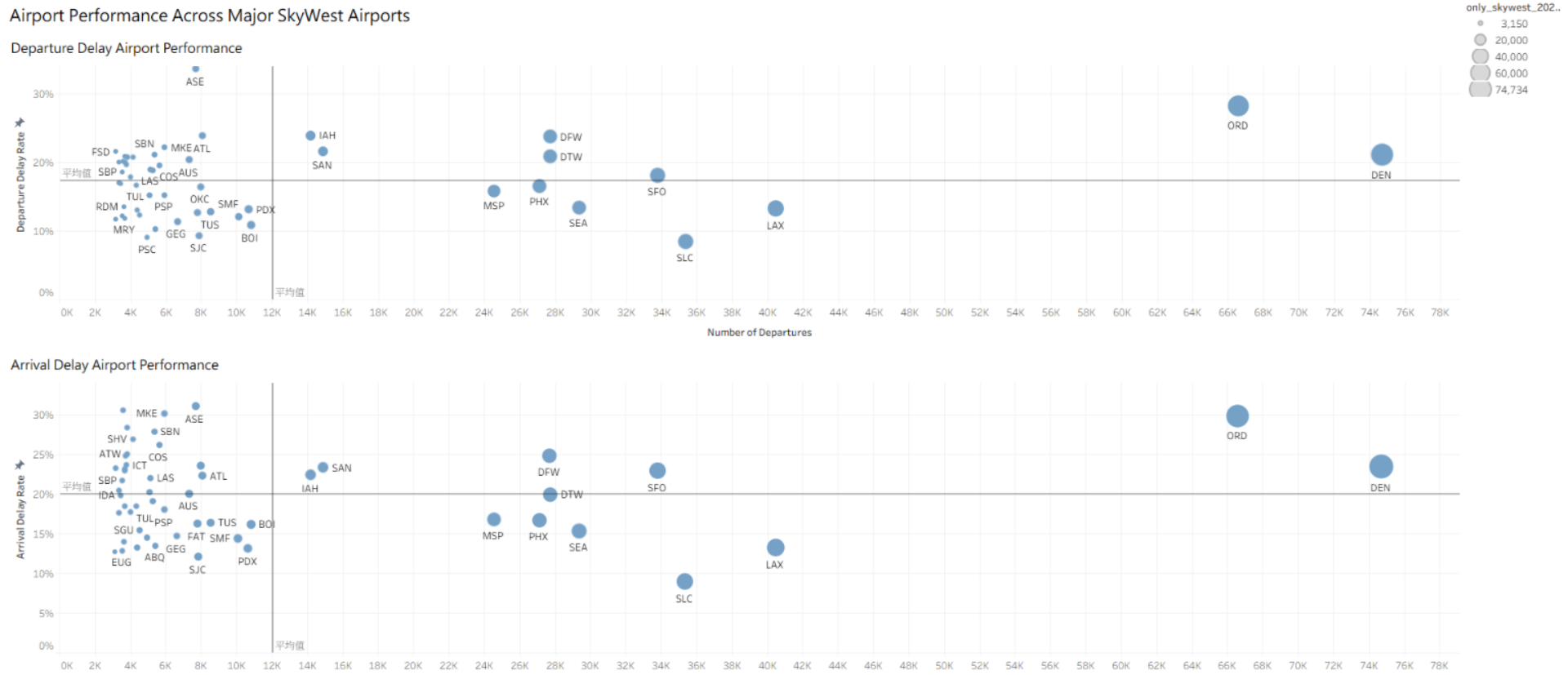


Departure and arrival delay rates follow a similar monthly pattern. December records the highest delay levels, while September shows the lowest rates.

Delay Heatmap

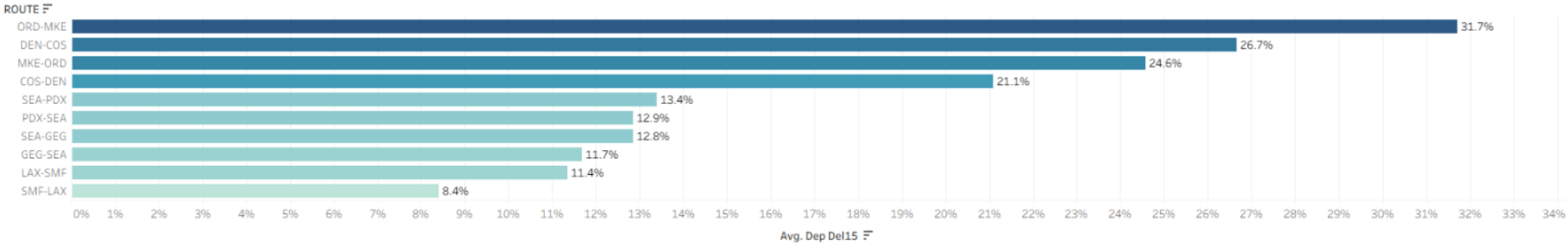


Airport Performance Across Major SkyWest Airports

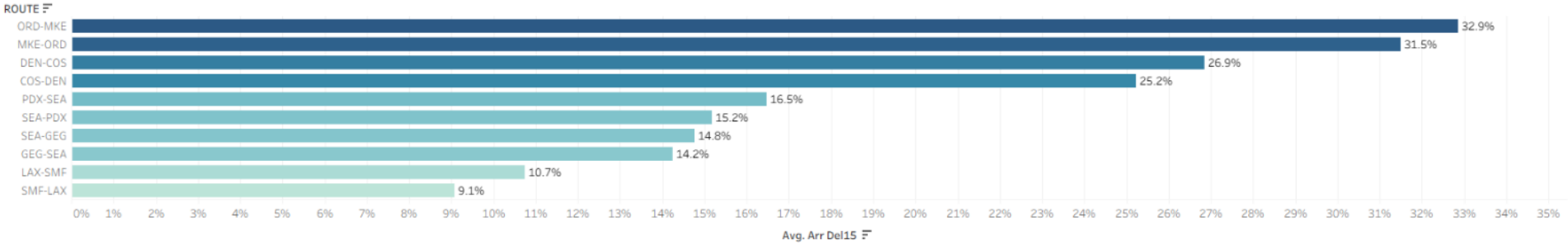


Departure vs. Arrival Delay Rates

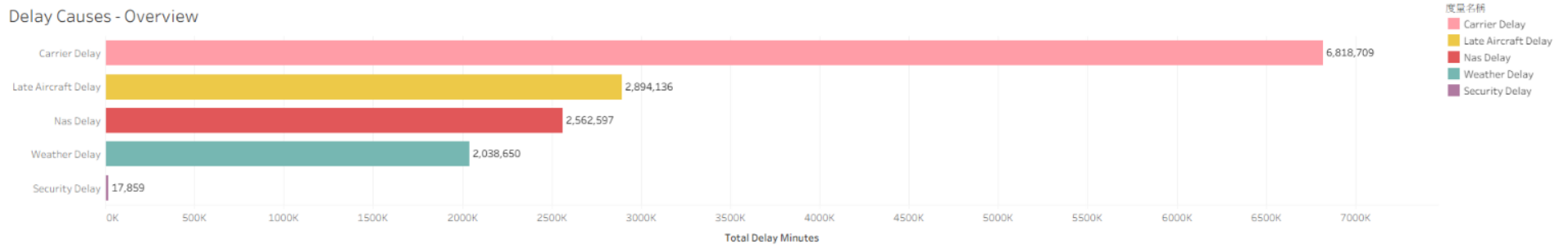
Departure Delay Rate by Route - Top 10



Arrival Delay Rate by Route - Top 10

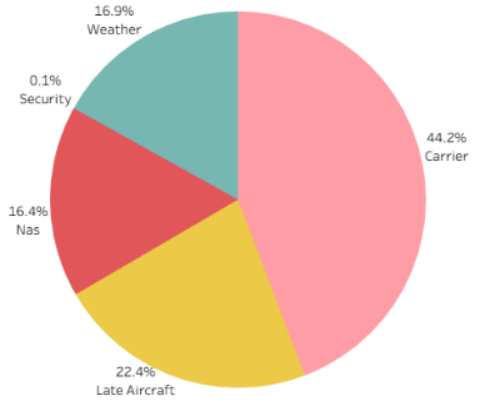


Delay Causes - Overview

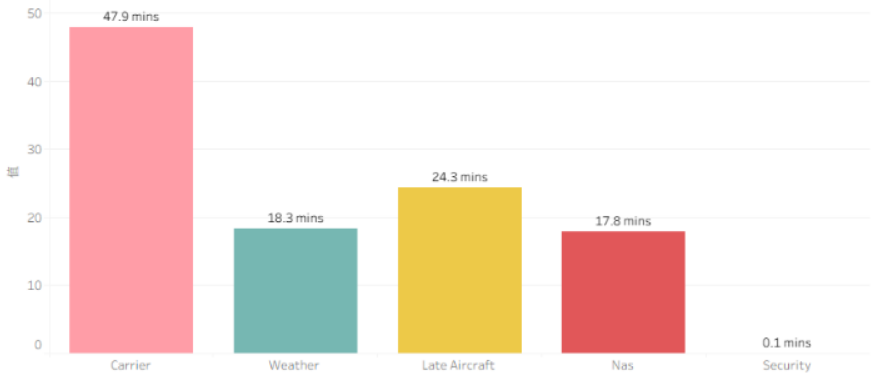


Key Takeaway: Carrier delay has the greatest overall impact and remains the largest delay cause in December. It also has the highest average severity at 47.9 minutes, showing that carrier-related issues are important in both frequency of impact and delay length.

Highest Impact Dec

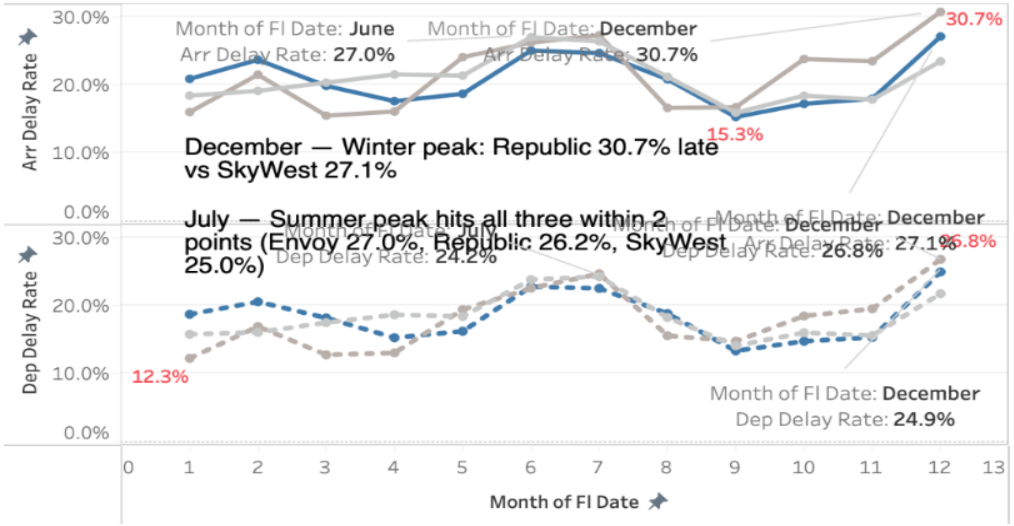


Delay Cause (Average Severity) - Dec

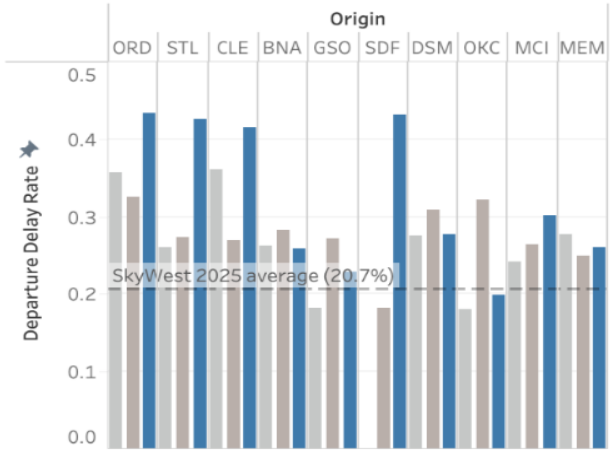


SkyWest vs Envoy and Republic - on-time performance, 2025

Monthly Delay Rate Trend



SkyWest vs competitors by airport - Departure Delay Rate, 2025



Carrier Name
- Envoy Air
- Republic Airways
- SkyWest

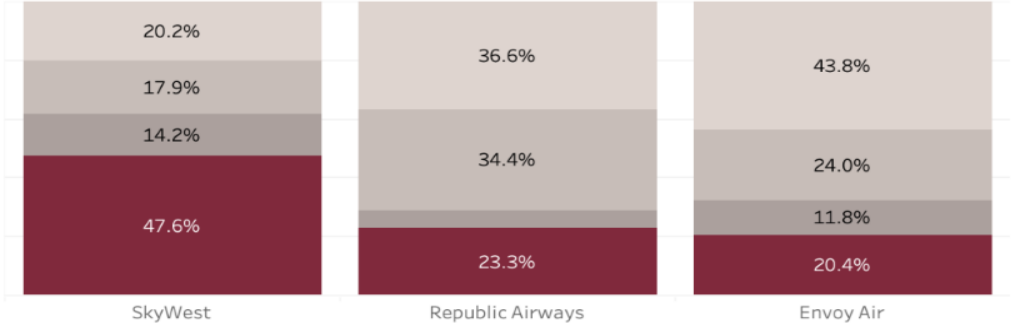
Measure Names
- Late Aircraft
- Nas
- Weather
- Carrier

Select Metric: Departure Delay Rate

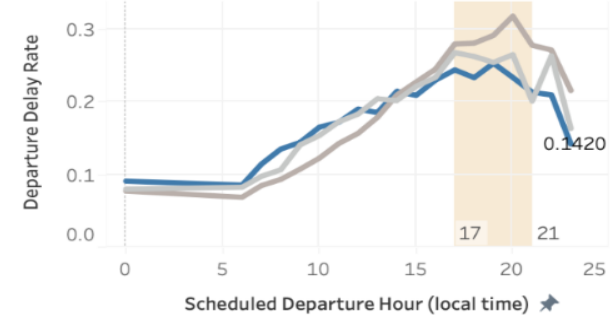
Month of Fl Date: December

Show history

Carrier-caused delay is 47.6% of SkyWest's delay minutes - more than double either competitor



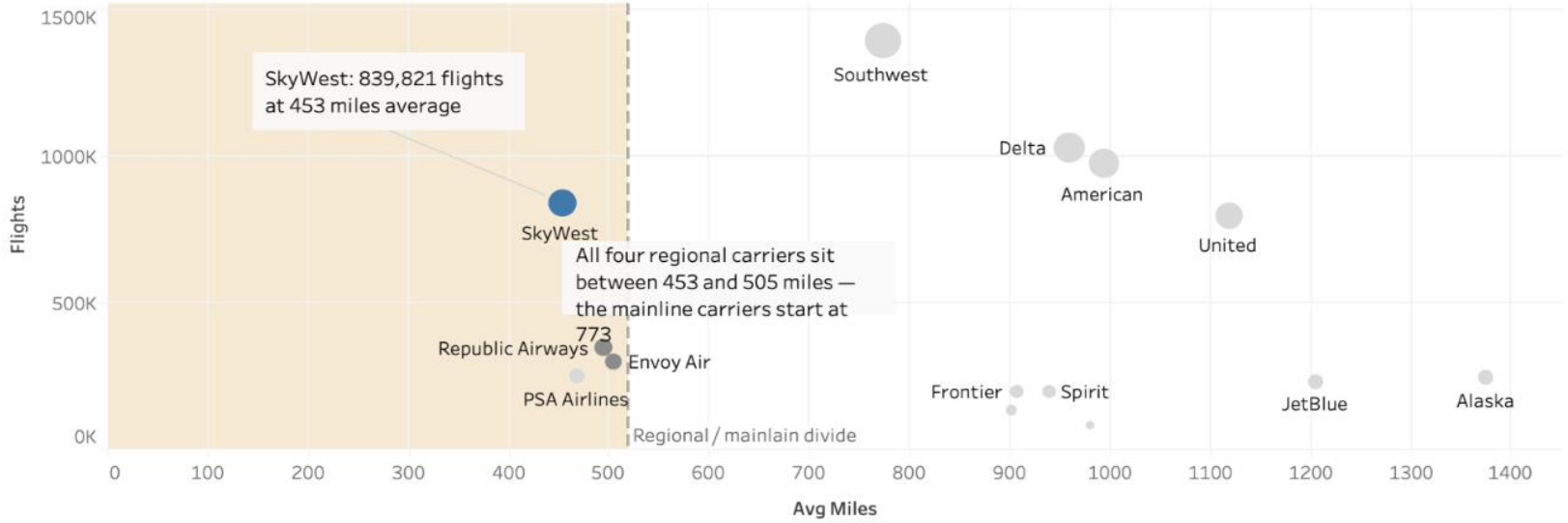
SkyWest vs competitors through the day - Departure Delay Rate, 2025



Why Envoy and Republic?

Select Metric: Departure Delay Rate

Regional carriers cluster below 520 miles per flight - SkyWest's peers are Envoy and Republic



Comparison Scorecard - SkyWest cancels least and flies latest

Carrier Na..	Dep Delay Rate	Arr Delay Rate	Delay Propagation	Avg Dep Delay (min)	Avg Arr Delay (min)	Cancellation Rate	Delay Min per flight
SkyWest	18.5%	20.7%	+2.3%	17.2	18.3	1.4%	17.0
Republic Airways	18.1%	21.5%	+3.4%	But 2.2 more delay minutes per flight than Republic	16.3	Less than half Republic's cancellation rate	14.9
Envoy Air	18.5%	21.0%	+2.6%	12.8	14.3	2.2%	13.0

Thank You

Q & A